

NIH BIOGRAPHICAL SKETCH COMMON FORM

Name: Bisarad, Pritha

Persistent Identifier (PID) of the Senior/Key Person: <https://orcid.org/0000-0002-2677-9847>

Position Title: Research Scientist

Organization and Location: Phoenix Children's, Department of Neurology, Phoenix, Arizona, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Arizona State University, Tempe, Arizona, United States	PhD (Chemistry)	07/2016	08/2022	Chemistry/Computational Biology
University of Calcutta, Kolkata, West Bengal, IN	M.Sc	06/2013	06/2015	Chemistry
University of Calcutta, Kolkata, West Bengal, IN	B.Sc	08/2010	05/2013	Chemistry

Appointments and Positions

2026 - present	Research Scientist, Phoenix Children's, Department of Neurology, Phoenix, Arizona, United States
2022 - 2025	Postdoctoral Fellow, Phoenix Children's, Department of Neurology, Phoenix, Arizona, United States
2021 - 2022	President, Biophysical Society Arizona Student Chapter, Tempe, Arizona, United States
2019 - 2021	Treasurer, Biophysical Society Arizona Student Chapter, Tempe, Arizona, United States
2017 - 2018	Research Collaborator, HealthTell Inc (iCarbonX), Chandler, Arizona, United States
2016 - 2020	Graduate Teaching Assistant, Arizona State University, Tempe, Arizona, United States
2015 - 2015	Research Intern, TCG Lifesciences, Kolkata, Not Applicable, N/A, India

Products**Products Closely Related to the Proposed Project**

1. Pascual M, Bisarad P, Kelbert J, Chinander S, Gelineau-Morel R, Gorodetsky C, Hewitt A, Larsh T, Lucente N, O'Malley J, van der Werf L, Hauptman J, Sanger T, Ponce F, Kruer M, Thompson J. Stimulation-related increases in power spectral density covary with clinical evidence of overstimulation during deep brain stimulation for pediatric dystonia. Neuromodulation: Technology at the Neural Interface. 2026 May 25. Available from: <https://www.sciencedirect.com/science/article/pii/S1094715926001236> Other: 1094-7159
2. Kelbert J, Guest A, Bisarad P, Larsh T, Bhatia P, Chinander S, Cornejo P, van der Werf L, Ponce F, Thompson J, Kruer M. Local Field Potential-Based Programming for Deep Brain Stimulation in Pediatric *DYT1* Dystonia. Movement Disorders Clinical Practice. 2024 December 03; 12(2):249-252. Available from: <https://movementdisorders.onlinelibrary.wiley.com/doi/10.1002/mdc3.14283> DOI: 10.1002/mdc3.14283

Other Significant Products Highlighting Contributions to Science

1. Galaz-Montoya CI, Lewis SA, Galindo MK, Cornejo P, Skidmore PT, Bisarad P, Magee H, Bontempo K, Keren B, Afenjar A, Skorvanek M, Zech M, Wentzensen IM, Gurnett CA, Chung WK, Bakhtiari S, Kruer MC. Expanding the Phenotypic Spectrum of the Recurrent De Novo FBXO31 p.Asp334Asn Variant: Evidence for a Novel Neurodevelopmental Disorder (Kruer Syndrome). Clin Genet. 2026 Aug;110(2):227-235. PubMed PMID: [41858232](#).
2. Bisarad P, Wang Y, Skidmore P, Galaz-Montoya C, Lewis S, Alhaddad B, Kong N, Julian D, Magee H, Kruer T, Xie Y, Zheng W, Li B, Rajabpour F, Liu J, Revanur A, Bakur K, Firouzabadi S, Sharbatkhori S, Tafakhori A, Taghiabadi E, Nezaminargabad E, Vosoogh S, Jamshidi J, Arefnia S, Hosseini S, Khajehmirzaei A, Jamali F, Ahmadifard A, Khodadadi H, Daneshmand P, Bohlega S, Maddirevula S, Nadeef S, Hashem M, Salih M, Mohamed I, Sticht H, Morias S, Damásio J, Santos M, Loureiro J, Rodrigues R, Stevanin G, Benkirane M, Dauriat B, Head N, Baptista J, Shahhosseini S, Mohammad F, Zhao H, Padilla-Lopez S, Alkuraya F, Bakhtiari S, Kruer M, Jin S, Darvish H. Recessive genomic and phenotypic variation in consanguineous families with cerebral palsy. [Preprint]. 2025 November 06. DOI: 10.1101/2025.11.04.25339178
3. Bisarad P, Kelbauskas L, Singh A, Taguchi A, Trenchevska O, Woodbury N. Predicting monoclonal antibody binding sequences from a sparse sampling of all possible sequences. Communications Biology. 2024 August 12; 7(1):- . Available

from: <https://www.nature.com/articles/s42003-024-06650-3> DOI: 10.1038/s42003-024-06650-3

4. Nosrati MSS, Doustmohammadi A, Severino M, Romano F, Zafari M, Nemati AH, Velmans C, Netzer C, Breuer J, Broekaert JJ, Joachim A, Almasri N, Kruer MC, Skidmore P, Bisarad P, Hoque J, Bakhtiari S, Torella A, Nigro V, Buffelli F, Fulcheri E, Müller A, Zara F, Capra V, Scala M. Novel KIF26A variants associated with pediatric intestinal pseudo-obstruction (PIPO) and brain developmental defects. Clin Genet. 2025 Jan;107(1):83-90. PubMed Central PMCID: [PMC11608842](https://pubmed.ncbi.nlm.nih.gov/PMC11608842/).

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

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NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Bisarad, Pritha

Persistent Identifier (PID) of the Senior/Key Person: <https://orcid.org/0000-0002-2677-9847>

Position Title: Research Scientist

Organization and Location: Phoenix Children's, Department of Neurology, Phoenix, Arizona, United States

Personal Statement

I am a research scientist at Phoenix Children's and the University of Arizona whose research focuses on the molecular and electrophysiological mechanisms underlying neurological movement disorders. My academic training spans molecular biochemistry, omics sciences, immunology, electrophysiology, and computational biology. As a chemistry student, I developed a strong molecular perspective on physiological mechanisms that continues to inform my research today.

During my PhD with Dr. Neal Woodbury, I investigated the sequence-dependent nature of protein-target interactions, particularly antibody-antigen binding. This work provided rigorous training in computational biology, biostatistics, and machine learning. As a doctoral candidate, I developed and optimized a deep neural network to predict sequence-binding relationships between monoclonal antibodies and target peptides (Bisarad et al., 2024). By integrating large-scale microarray-derived omics datasets with machine learning approaches, I quantified the sequence-dependent nature of biomolecular interactions, resulting in work that received recognition at national conferences. I also developed novel experimental protocols to characterize monoclonal antibody binding profiles on peptide microarrays through a collaboration with HealthTell. In addition, I received a teaching award for mentoring undergraduate students.

For my postdoctoral training, I expanded my research interests to the genetic and neurophysiological basis of movement disorders, allowing me to apply my expertise in machine learning, biostatistics, and multi-omics analyses to clinically relevant neurological questions. Under the mentorship of Dr. Michael Kruer, whose work has advanced deep brain stimulation (DBS) treatment and neurophysiological characterization of dystonia and chorea, I have focused on identifying objective physiological measures of disease. Despite significant advances in DBS, clinicians still lack reliable biomarkers to quantify disease severity, track symptom fluctuations, and guide therapeutic optimization in movement disorders. Local field potentials (LFPs) provide a unique opportunity to directly interrogate pathological neural circuit activity in vivo and may offer biologically grounded measures of disease state and treatment response.

Through my research endeavors, I seek to advance the use of LFPs as objective indicators of disease activity and treatment response in dystonia and chorea, enabling more precise neuromodulation strategies and improving patient outcomes (Pascual et al., 2026, Kelbert et al., 2024). My goal is to develop clinically meaningful neurophysiological markers that better characterize disease states, track symptom severity, and guide therapeutic interventions. By integrating clinical observations with neural recordings obtained from implanted devices, I aim to uncover the circuit-level mechanisms underlying pathological movement patterns. Therefore, as part of our research team, I seek to: (a) identify and validate electrophysiological signatures within LFP recordings that correlate with symptom severity and longitudinal clinical outcomes in pediatric movement disorders, and (b) develop data-driven frameworks that leverage neural recordings to guide personalized DBS programming and inform future adaptive neuromodulation strategies.

Honors

2025	Reviewer's Choice Abstract, American Society of Human Genetics
2025 - 2027	Clinical Research Training Scholarship, American Academy of Neurology
2020	Distinguished Teaching Assistant Award, Arizona State University
2013 - 2015	Indira Gandhi Scholarship, University Grant Commission, India

Contributions to Science**1. Characterizing movement disorders using local field potentials recorded from deep brain stimulation devices**

A major challenge in administering deep brain stimulation (DBS) successfully in movement disorders like dystonia and chorea is the lack of objective physiological measures to quantify symptom severity and guide therapy. Local field potentials (LFPs) recorded from implanted deep brain stimulation (DBS) devices provide a unique opportunity to directly assess pathological neural circuit activity.

My research has contributed to establishing the clinical relevance of LFPs in movement disorders. We demonstrated that longitudinal changes in LFP activity tracked clinical improvement in DYT1-associated dystonia symptoms following successful DBS therapy, supporting the utility of LFPs as indicators of treatment response (Kelbert et al., 2024). We further showed that excessive DBS stimulation can worsen dystonia symptoms and is accompanied by corresponding changes in LFP activity (Pascual et al., 2026). Building on these findings, I have been characterizing the electrophysiological signatures in larger cohorts of patients with dystonia and chorea and evaluated relationships between LFP activity, symptom severity, and clinical outcomes.

This work supports the development of objective neurophysiological measures for movement disorders and may facilitate personalized DBS programming and future adaptive neuromodulation strategies.

My specific contributions included electrophysiological data analysis, development of computational pipelines, integration of clinical and neural recording datasets, statistical analyses, and interpretation of findings.

2. Using computational genomics to identify genetic causes and mechanisms of neurodevelopmental movement disorders

The genetic and molecular basis of many neurodevelopmental movement disorders remains poorly understood, particularly in underrepresented and consanguineous populations where recessive disease mechanisms are enriched. My research focuses on developing and applying computational approaches to identify disease-causing variants and investigate the biological pathways underlying cerebral palsy, dystonia, and related neurological disorders.

A major contribution of my work has been the characterization of recessive genomic variation in consanguineous families with cerebral palsy, providing new insights into the genetic architecture of this heterogeneous disorder (Bisarad et al., 2026 (preprint)). I have also contributed to the characterization of FBXO31-related neurodevelopmental disease and the identification of novel KIF26A variants associated with brain developmental abnormalities, further expanding the spectrum of genes implicated in childhood neurological disease (Galaz-Montoya et al., 2026; Nosrati et al., 2025). To facilitate gene discovery, I developed computational approaches for variant prioritization in consanguineous pedigrees and contributed to bioinformatics pipelines supporting genomic analyses of more than 500 families affected by neurodevelopmental movement disorders.

This work advances molecular diagnosis, improves understanding of disease mechanisms, and provides computational frameworks for identifying novel genetic contributors to neurological disease.

My specific contributions included genomic data analysis, development of computational and bioinformatics pipelines, variant prioritization, genotype-phenotype correlation, network and pathway analyses, and integration of large-scale clinical and genomic datasets.

3. Using machine learning to predict antibody sequence-binding relationships

Understanding the molecular determinants of antibody-epitope recognition remains a fundamental challenge in immunology. Because experimentally characterizing antibody binding across the vast peptide sequence space is impractical, computational approaches are needed to predict binding behavior from limited experimental data.

My research demonstrated that machine learning can infer high-specificity antibody-epitope interactions from low-specificity interactions with random peptide libraries. I developed a deep neural network trained on experimentally measured interactions between monoclonal antibodies and random nonamer peptides sparsely sampled from sequence space. The resulting models predicted antibody binding patterns across the entire nonamer sequence space without prior knowledge of target epitopes, providing new insights into the sequence-dependent nature of molecular recognition.

This work advances the application of neural networks to molecular recognition by providing a scalable approach for characterizing antibody specificity, with potential applications in therapeutic development and epitope discovery. My specific contributions included developing the deep learning framework, performing computational analyses, integrating peptide microarray datasets, interpreting sequence-binding relationships, and developing complementary experimental methods for antibody characterization. This work resulted in a peer-reviewed publication (Bisarad et al., 2024) and a US patent (pending) for technological innovation.

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